

Junlin Wang

junlin.wang@wustl.edu | GitHub | LinkedIn

RESEARCH INTERESTS

Efficient large language models: understanding how capacity is distributed across a network's layers and modules, and using that structure—often through spectral (heavy-tailed) analysis—to guide compression, parameter-efficient adaptation, and knowledge editing. And interest in AI for health: faithful and interpretable clinical language models, and multimodal analysis of clinical notes and medical imaging.

EDUCATION

Washington University in St. Louis

St. Louis, MO, USA

M.S. in Data Analytics and Statistics— GPA: 3.61/4.0

Jan 2025 – Dec 2026

- **Selected coursework:** Bayesian Methods in Machine Learning (A), Deep Reinforcement Learning, Introduction to Machine Learning (A), Introduction to Artificial Intelligence (A–), Probability & Stochastic Processes (A+), Optimization (A), Statistical Computation (A), Analysis of Algorithms, Data Structures & Algorithms (A).

Wuhan University

Wuhan, China

B.S. in Financial Engineering — GPA: 3.5/4.0

Sep 2020 – May 2024

- **Coursework:** Probability & Mathematical Statistics, Machine Learning, Digital Image Processing, Data Analysis in R, Database Systems, Network Programming, C Programming.
- **Honors:** Outstanding Student Scholarship; Third-Class Scholarship; Outstanding Student in Social Activities.

RESEARCH EXPERIENCE

Research Assistant — LLM Pruning & Efficient Adaptation

Remote

Dartmouth College — Advisors: Prof. Yaoqing Yang, Prof. Hengrui Luo from Rice university

Sep 2025 – Present

- Led a 3,000+-run study of sparsity allocation for LLM pruning across seven models (LLaMA-2/3.1, Qwen-2.5, Mistral, OPT), showing that allocating pruning budgets by a weight matrix's **position** (module type $\times$ depth) consistently **outperforms** allocating from raw per-matrix measurements; beats five allocation baselines on all seven models at 80% sparsity. And this is my paper under writing right now
- Found gains from layer decisions guided by the **module $\times$ depth decomposition** of weight spectra—its structural component and its residual—across multiple settings: sparsity allocation for pruning (the study above); **fine-tuning layer selection** on LLaMA-family models, where *residual*-selected layers win 8 of 11 depth-matched comparisons; and **LoRA rank allocation**, where *position-structured* rank maps outperform faithful reproductions of AlphaLoRA by 0.9–1.6 pp on GSM8K.
- Investigated the **mechanism of the residual** component of layer-wise spectral structure through three probes: a training-checkpoint time-course on Pythia-1.4B/2.8B tracking when module, depth, and residual structure each emerge during training; regression analyses testing whether a layer's residual predicts its subsequent training dynamics; and cross-architecture fine-tuning probes (LLaMA vs. Qwen) testing residual-guided layer selection.
- Found that editing and fine-tuning methods imprint **signature spectral footprints** on this residual, ordered by how regularized their updates are: ROME $\rightarrow$ pure residual (Kendall $\tau=1.0$, 34/34), MEMIT $\rightarrow$ nearly pure depth ($\rho \approx 0.9$), AlphaEdit $\rightarrow$ near-flat—enabling spectra-based auditing of how a model was modified; the induced layer-selection rules are **architecture-dependent** (the best fine-tuning targets on LLaMA are the worst on Qwen).
- Building a pipeline for **layer-wise method selection**: comparing knowledge editing (ROME, MEMIT, AlphaEdit) with parameter-efficient fine-tuning PEFT (LoRA, DoRA, QLoRA) for factual updating—TriviaQA reformatted into an editing benchmark with LLM-generated paraphrases, scoring reliability, generalization, and locality under sequential and batched editing, with spectral layer-quality metrics as the selector.

Graduate Research Assistant — AI for Health Lab

St. Louis, MO

Washington University in St. Louis — Advisor: Prof. Chenyang Lu

Dec 2025 – Present

- Built a **sentence-level rationale attribution** framework for clinical notes in degenerative cervical myelopathy (DCM) on 3 prediction heads, evaluating whether gradient-based attributions identify the sentences that actually support a model's prediction. And compare gradient methods to Attention method, Directly generate method and occlusion sentence method. Finally we find Gradient method has better robustness and recall performance.

- Benchmarked **six model families** (ModernBERT, PubMedBERT, BioGPT, LLaMA-3.2-1B, LLaMA-3.1-8B, Meditron-7B) against GPT-5-generated gold-standard annotations; ModernBERT and LLaMA-3.2-1B aligned best with gold rationales ($R = 0.52$ and 0.54), beating the domain-pretrained baselines.
- Implemented LLM-guided HPO (GEPA: reflective-LM, Pareto-based) and compared it against Weights & Biases Bayesian sweeps on seven backbones; fixed-seed reruns reproduced validation AUROC yet selected different hyperparameters in 6/6 models, revealing a flat optimization landscape in which the GEPA-vs-sweep gap falls within reproduction noise, verified with an EHR-only negative control.
- Implemented the LoRA + bf16 training stack on an HPC cluster under a **HIPAA-compliant** workflow.

Research Assistant — IMAGINE Project

St. Louis, MO

Mallinckrodt Institute of Radiology, WashU Medicine — Supervisors: Mark Anastasio, Paul Commean Aug 2026 – Jun 2027

- Developing AI-based analysis of pediatric skeletal ultrasound for the IMAGINE project (Integrated Mobile AI technology for Global Infant Nutrition Evaluation) at WashU's Mallinckrodt Institute of Radiology — a multi-site international effort to extract skeletal and neurodevelopmental biomarkers from portable ultrasound for low-cost, point-of-care nutritional assessment — with a focus on deep-learning ossification scoring of infant hand/wrist bones as a bone-age marker of early malnutrition.
- Built an end-to-end bone-surface segmentation pipeline for pediatric hand/wrist ultrasound DICOM images using localized active contours (Lankton & Tannenbaum, 2008; Martins et al., EMBC 2016), with template-based contour initialization, shadow-aware near-field rejection, and automated QC overlays and parameter sweeps for clinician review.

MANUSCRIPTS

In preparation / under submission Status as of September 2026.

1. **Junlin Wang, other coauthor**, “Full-Position Pruning: Allocating Sparsity by Module and Depth in Large Language Models.” *Manuscript in preparation.*
2. **Junlin Wang, other coauthor**, “Sentence-Level Rationale Attribution in Clinical Notes.” *In preparation.*

TEACHING EXPERIENCE

Graduate Teaching Assistant — ESE 5200: Probability & Stochastic Processes

St. Louis, MO

Washington University in St. Louis

Jan 2026 – May 2026

- Ran weekly office hours and exam review sessions on probability theory, Markov chains, and stochastic processes; graded problem sets and exams for a graduate-level cohort of 40 students.

Graduate Teaching Assistant — CSE 4107: Machine Learning

St. Louis, MO

Washington University in St. Louis

Sep 2025 – Dec 2025

- Held weekly office hours on supervised learning, optimization, and neural network fundamentals; graded assignments and final projects, and advised students on course project design and evaluation.

SELECTED PROJECTS

Bayesian Optimization with Gaussian Process Surrogates

Course project, *Bayesian Methods in Machine Learning* [github address]

Fall 2025

- Implemented Bayesian optimization with GP surrogates and Expected Improvement acquisition; performed GP model selection via BIC; evaluated on Branin and on SVM / online-LDA hyperparameter tuning, quantifying speedup over random search across repeated randomized trials.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL, R, $\text{\LaTeX}$

ML / DL: PyTorch, Hugging Face Transformers, PEFT (LoRA/RoSA), scikit-learn, NumPy, pandas

Methods: fine-tuning & parameter-efficient adaptation, structured pruning and low-rank compression, knowledge editing, mixed-precision (bf16) and distributed training, attribution & interpretability, benchmarking and evaluation design

Infrastructure: Weights & Biases, Git, Linux/bash, SLURM/HPC clusters